Endogenizing Nature in Production Theory: A Framework of Economic Commensalism

Fernando Souza da Vieira[1] Alexandre Chibebe Nicolella[2]

[1]Faculty of Economics, Administration and Accounting of Ribeirão Preto, University of São Paulo, Brazil

Abstract

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation.

Problem

Keywords: economic commensalism; environmental economics; rational incentives; commons; carbon markets.

1. Introduction

The relationship between economic activity and the natural environment has long been marked by a persistent tension between productive expansion and ecological preservation. The environment is essential to human welfare and economic production, yet the depletion of natural resources and the degradation of ecological systems can undermine the very productive capacity that economic activity seeks to expand. Conversely, production, investment, and technological progress generate employment, income, and improvements in living standards, so environmental policy cannot be evaluated independently of its effects on private incentives and productive viability. The relevant economic problem is therefore not whether production or preservation should prevail, but under what conditions the use of nature can remain compatible with the ecological functions on which present and future production depend.

Economic theory does not ignore nature. Land, natural resources, environmental costs, externalities, and common-pool resources have been extensively examined across different strands of the literature, such as Pigou (2010), Coase (1960), Hardin (1968), Ostrom (1990), and Dasgupta (2001), as we will see in detail in Section 3.

[1] fernandodvieira@usp.br
[2] nicolella@usp.br

In canonical representations of production, such as the Cobb–Douglas function, output is usually modeled as a function of capital and labor, sometimes with land or natural resources appearing as inputs, but ecological systems are not generally represented as a factor or cofactor that affects output, future productivity, strategic incentives, and intertemporal welfare. Instead, the environment often appears as a stock, a constraint, a source of externalities, or an object of corrective regulation, and this representation has economic consequences. This modeling choice is not neutral: by representing the environment primarily as a stock, constraint, or externality, standard frameworks limit the extent to which ecological functionality can enter private decision problems as a state that co-determines productivity.

When ecological functionality remains outside the private decision problem, preservation tends to appear as a legal obligation, a compliance cost, or an opportunity cost imposed on production. Its benefits are frequently delayed, uncertain, spatially diffuse, or collectively appropriated, whereas the returns from resource conversion may be immediate, measurable, and privately captured. Even an agent who recognizes the long-term importance of ecological integrity may therefore rationally choose short-term expansion when the expected private return from degradation exceeds the expected private return from preservation. The divergence is intensified by credit constraints, competitive pressures, incomplete information, weak enforcement, and uncertainty regarding future policies.

Governmental command-and-control regulation and penalties, carbon markets, environmental certification, payments for ecosystem services, and fiscal incentives translate parts of these insights into policy. These approaches have generated important improvements in specific dimensions, but they address different components of the problem and do not automatically establish a unified relationship among ecological conditions, production, private payoffs, and territorially grounded governance.

Nevertheless, the remaining gap is neither the absence of nature from economic thought nor the absence of legal, financial, institutional, or normative instruments intended to protect it. Rather, what remains insufficiently specified is an integrated set of conditions under which ecological functionality is jointly determined with production, its productive and ecosystem-service effects enter private payoffs, and governance is matched to the spatial scale of the relevant ecological processes. The gap addressed in this article is therefore one of analytical integration rather than the absence of environmental considerations from economics.

Against this background, the article seeks to resolve the following **problem**: under what conditions can ecological functionality be represented as an endogenous state of production and institutionally governed so that its maintenance becomes incentive-

compatible with private productive viability? This question replaces a simple opposition between use and preservation with a problem of institutional and productive design. It also shifts the analysis from whether economic activity causes any environmental impact to whether it systematically erodes the ecological conditions required for future production and social welfare.

The **general objective** of the article is to develop a theoretical-analytical framework that represents ecological functionality as an endogenous state of production and derives the conditions under which territorially aligned governance and private incentives can make its maintenance compatible with productive viability. On this basis, the article defines and characterizes economic commensalism as a distinct ecological-economic configuration.

More specifically, the **specific goals** are: (i) to distinguish the use of natural resources as conventional inputs from the role of ecological functionality as a state that affects production and responds to productive and preservation decisions; (ii) to examine how the spatial heterogeneity and imperfect substitutability of ecosystem services affect the appropriate scale of environmental governance and compensation; (iii) to derive the payoff conditions under which maintaining ecological functionality becomes individually rational for productive agents; (iv) to define economic commensalism formally and compare its productive and ecological implications with predatory, extractive, and mutualistic resource-use regimes.

The analysis is guided by the following **general hypothesis**: When ecological functionality jointly determines productive outcomes and is itself affected by resource-use and preservation decisions, institutional arrangements that incorporate locally relevant ecosystem benefits and costs into private payoffs can admit a commensal configuration in which productive activity remains privately viable without eroding the ecological functions on which it depends.

This general hypothesis is decomposed into three **specific hypotheses**.

(H1) Productive-ecological endogeneity. When ecological functionality affects output and simultaneously responds to production, preservation, and institutional rules, treating it as a fixed input or exclusively as an external effect omits feedback mechanisms that alter marginal returns and may change the relative ranking of alternative resource-use strategies.

(H2) Local governance. When ecosystem functions are spatially heterogeneous and imperfectly substitutable, governance and compensation mechanisms anchored in ecologically relevant territorial units can reduce the divergence between private and territorial returns to preservation more effectively than spatially fungible mechanisms, provided

that relevant ecological effects are observable, local institutions possess sufficient implementation capacity, and cross-boundary spillovers are coordinated at broader institutional levels.

(H3) Intertemporal incentive compatibility. When private payoffs incorporate the contribution of ecological functionality to current and expected future production, remunerated ecosystem services, and avoided losses from degradation, the range of opportunity costs for which preservation is privately optimal becomes larger than under a payoff based exclusively on current marketed output. Preservation becomes individually rational whenever the resulting marginal private benefits equal or exceed its opportunity and implementation costs.

These hypotheses are analytical rather than directly empirical. H1 is examined by comparing the marginal conditions obtained under exogenous and endogenous representations of the ecological state; H2 by comparing alternative spatial and institutional configurations; and H3 by deriving the threshold at which the private return to maintaining ecological functionality becomes at least as large as the return to degradation. The dynamic stability of this configuration under uncertainty, learning, and repeated decisions is left to the subsequent intertemporal model.

The article makes three principal contributions. First, it distinguishes the inclusion of natural resources as conventional inputs from the endogeneity of ecological functionality as a jointly determined state of production. Second, it connects this productive representation to private payoffs and to governance arrangements anchored in ecologically relevant territorial units. Third, it defines economic commensalism through the joint behavior of private viability and ecological functionality and derives the conditions under which this configuration can emerge relative to predatory, extractive, and mutualistic regimes. Together, these contributions provide a conceptual and analytical basis for treating preservation not only as an external constraint, but as a potentially incentive-compatible component of productive strategies.

Methodologically, the article adopts a theoretical-analytical approach that combines production theory, environmental economics, and institutional analysis. It first reconstructs how externalities, property rights, common-pool resources, and natural capital illuminate distinct parts of the production-preservation problem. It then develops a preliminary model in which the environmental state enters the production structure and the payoff of productive agents, while incentives and governance affect the relative return to preservation and exploitation. The model is used to compare alternative ecological-economic strategies and to derive the conditions under which a commensal configuration can emerge as an individually rational and ecologically non-degrading outcome. The anal-

ysis remains deliberately foundational: it establishes the conceptual and static analytical basis upon which intertemporal dynamics, uncertainty, learning, and empirical testing can subsequently be developed.

The remainder of the article is organized as follows. Section 2 contextualizes the production-preservation problem and the limitations of current environmental governance. Section 3 reviews the relevant literature, moving from externalities and property rights to common-pool governance and natural capital, and identifies the elements that remain analytically disconnected. Section 4 develops the conceptual and analytical model of economic commensalism, defines the alternative ecological-economic regimes, and states the relevant equilibrium conditions. Section 5 presents the theoretical results and comparative implications of the model. Section 6 discusses their relationship to the existing literature, their institutional meaning, and the limitations of the proposed framework. Section 7 concludes and indicates directions for dynamic formalization and empirical assessment.

2. Problem Context

The production-preservation problem unfolds in a context characterized not by institutional absence, but by an increasingly dense architecture of environmental governance. Over the past three decades, international agreements, national regulations, carbon-pricing systems, payments for environmental services, financial instruments, certification schemes, and supply-chain commitments have sought to constrain degradation or redirect economic activity toward less environmentally damaging outcomes. These mechanisms differ in legal status, object, territorial reach, and theory of change. Some establish aggregate objectives; others prohibit specific conduct, price measurable externalities, compensate observable conservation, condition access to capital, or disclose environmental attributes to consumers and investors. Their coexistence therefore does not constitute a single governance system. It constitutes a policy mix whose components intervene at different margins of the production-preservation problem.

This institutional expansion has occurred alongside environmental trends that remain inconsistent with stated collective objectives. The synthesis of the Sixth Assessment Report documents widespread climate-related impacts and a narrowing set of feasible pathways for limiting additional warming (IPCC, 2023). The global biodiversity assessment, in turn, estimates that around one million species are threatened with extinction and that human activity has significantly altered approximately three quarters of the terrestrial environment (IPBES, 2019). More recently, the *Emissions Gap Report 2025* estimated that full implementation of the available nationally determined contributions

would still imply warming of approximately 2.3–2.5°C over this century, while policies then in place implied approximately 2.8°C (UNEP, 2025). In Brazil, official monitoring reported that approximately 729,000 km^2, or 17% of the Amazon biome, had already been deforested by 2024 (INPE, 2024). These aggregate indicators coexist with periods of substantial improvement, including the sharp decline in Brazilian Amazon deforestation after 2004, which the literature associates with a combination of enforcement, protected areas, credit restrictions, and interventions in beef and soy supply chains (NEPSTAD et al., 2014). Environmental deterioration is therefore persistent and heterogeneous, but neither uniform nor institutionally invariant.

The coexistence of environmental degradation and environmental policy does not, by itself, demonstrate that existing instruments are ineffective. Such an inference would ignore the relevant counterfactual: degradation may have been greater in their absence, and individual instruments may produce substantial effects even when aggregate outcomes remain insufficient relative to collective goals. It is equally important not to collapse distinct evaluative concepts. Resource conversion may be privately rational when its expected private return exceeds the return to preservation; socially inefficient when relevant external costs remain unaccounted for; ecologically unsustainable when it erodes the functions required to maintain the system through time; or illegal when it violates an enforceable rule. These categories may overlap, but they are not equivalent. The problem examined here is consequently not reducible to misconduct, deficient ecological awareness, or an inherent moral opposition between private agents and collective welfare. It concerns the conditions under which individually rational decisions, interacting through markets and institutions, can generate outcomes that are collectively costly and ecologically degrading.

International environmental agreements address part of this problem by establishing common objectives, reporting procedures, and mechanisms for cooperation. The Kyoto Protocol imposed quantified mitigation commitments primarily on developed economies and introduced emissions trading, joint implementation, and the Clean Development Mechanism (UNFCCC, 1997). The Paris Agreement broadened participation through nationally determined contributions, successive cycles of ambition, transparency arrangements, and provisions for finance, technology, and capacity building (UNFCCC, 2015). The Convention on Biological Diversity established a parallel framework centered on conservation, the sustainable use of biodiversity, and the fair and equitable sharing of benefits derived from genetic resources (United Nations, 1992). These agreements are institutionally significant because climate stability and biodiversity conservation cannot be supplied through isolated national action alone. At the same time, their effectiveness depends on

the ambition of collective commitments and on their translation into domestic law, budgets, administrative capacity, and credible implementation. International coordination can define objectives and reduce informational and strategic barriers among states, but it does not directly determine the land-use or investment decision of each productive agent.

At the domestic level, command-and-control regulation translates environmental objectives into legally defined duties, standards, authorizations, and sanctions. The Brazilian Forest Code illustrates that this architecture is more heterogeneous than a simple opposition between prohibition and production. Law No. 12,651/2012 regulates Permanent Preservation Areas and Legal Reserves, establishes the Rural Environmental Registry and environmental regularization procedures, and provides for economic and financial mechanisms intended to support conservation and restoration (BRASIL, 2012). The law therefore combines property-specific restrictions, information and monitoring infrastructure, regularization mechanisms, and potential incentives. Scientific assessments of the reform similarly identify both reductions in restoration obligations and new opportunities for monitoring, compensation, and conservation (SOARES-FILHO et al., 2014; AZEVEDO et al., 2017). Characterizing the Forest Code exclusively as a rigid constraint would consequently overlook its hybrid design, just as characterizing it as an integrated solution would overlook the institutional conditions required to make its components operational.

Regulation can alter expected returns by reducing the legal set of available actions and by attaching an expected sanction to non-compliance. Its practical effect, however, depends on the probability of detection, the speed and credibility of enforcement, the clarity of obligations, and the capacity of regulated agents to comply. These conditions also shape distributional incidence. A conservation rule may be socially justified while imposing concentrated opportunity and implementation costs on landholders whose private returns from land conversion are more immediate than the benefits they can appropriate from maintained ecological functionality. This concentration of costs does not by itself establish injustice or inefficiency, but it can increase resistance and reduce compliance when obligations are uncertain, monitoring is inconsistent, or compensatory mechanisms are unavailable. In the Brazilian Amazon, simulations of combined enforcement and conservation payments reveal precisely this trade-off: adding payments may reduce landholder income losses, but it raises implementation costs and its cost-effectiveness varies spatially with deforestation risk, opportunity costs, and enforcement capacity (BÖRNER et al., 2015).

The distinction between formal rules and implemented rules is therefore essential. More than a decade after the 2012 reform, implementation of the Forest Code remains

heterogeneous across Brazilian states because general legal provisions must be translated, monitored, and enforced through multiple institutional layers. Recent institutional analysis identifies incomplete validation of self-declared information in the Rural Environmental Registry and uneven specification of state-level regularization procedures as central operational bottlenecks (OLIVEIRA; SCHNAIDER, 2025). Earlier evidence likewise showed that a land registry could lower monitoring costs without necessarily eliminating illegal deforestation when validation, enforcement, and compliance incentives remained incomplete (AZEVEDO et al., 2017). These findings do not imply that law is dispensable. They show that legal validity and practical effectiveness are different properties: the former defines obligations, while the latter depends on the institutions that connect those obligations to observable behavior and expected private payoffs.

Economic instruments intervene more directly in relative prices. Environmental taxes increase the private cost of polluting activities, while cap-and-trade systems constrain aggregate emissions and allow regulated agents to exchange allowances. In principle, both can induce abatement where its marginal private cost is lowest. In practice, instrument choice also involves distributional effects, uncertainty, administrative requirements, innovation incentives, and political feasibility; no instrument is unconditionally superior across all relevant criteria (GOULDER; PARRY, 2008). Empirical evidence should therefore be interpreted with similar care. The European Union Emissions Trading System reduced emissions relative to its estimated counterfactual despite initially low allowance prices (BAYER; AKLIN, 2020), and a systematic review of 80 ex-post evaluations found statistically significant reductions for at least 17 of the 21 carbon-pricing policies examined (DÖBBELING-HILDEBRANDT et al., 2024). Carbon pricing is thus neither generally ineffective nor sufficient by definition.

Its limitation for the present problem is primarily one of scope. A carbon price assigns a monetary consequence to an additional unit of measured greenhouse-gas emissions or to the use of a regulated allowance. It does not automatically represent biodiversity, hydrological regulation, soil integrity, ecological thresholds, or the productive feedbacks generated by their joint condition. Even within climate policy, a price may be too narrow to induce infrastructure, innovation, and institutional changes whose benefits are uncertain, complementary, or realized only over long horizons (ROSENBLOOM et al., 2020). The relevant conclusion is not that pricing should be abandoned, but that an instrument designed for a measurable externality cannot be presumed to govern the entire ecological state on which production and welfare depend.

Payments for environmental services approach the incentive problem from the opposite direction by remunerating an agreed conservation action or environmental outcome.

They can make preservation more attractive when payments are conditional, credible, and at least partly compensate opportunity and implementation costs. Reviews of the empirical literature find that such programs can generate additional conservation, while also emphasizing substantial variation in targeting, participation, monitoring, and estimated effects (BÖRNER et al., 2017; WUNDER et al., 2020). Their performance depends on whether the contracted outcome is additional to what would have occurred without payment, whether degradation is displaced elsewhere, whether environmental gains persist after the contract, and whether the relevant service can be measured at reasonable cost. Moreover, compensation for a specified service does not necessarily incorporate the broader contribution of ecological functionality to production. A payment for avoided carbon emissions, for example, may preserve carbon while remaining spatially indifferent to hydrological connectivity or biodiversity value unless these dimensions are explicitly included in program design.

Financial instruments operate at a different stage of the decision process. Green bonds, sustainability-linked finance, preferential credit, guarantees, and climate-related banking policies can influence the cost and availability of capital for environmentally designated activities. This channel matters because a profitable low-carbon or conservation investment may still fail to occur when it faces high upfront costs, long payback periods, credit constraints, or risks that private lenders cannot readily evaluate. For this reason, the financing problem cannot always be resolved by carbon pricing alone (CAMPIGLIO, 2016). Evidence on corporate green bonds indicates that labelled issuance can attract investor demand and is associated with subsequent environmental commitments and performance improvements, particularly when external certification strengthens credibility (FLAMMER, 2021). Yet a financing label identifies the use or conditions of capital rather than the ecological state of a territory. Its environmental integrity still depends on taxonomies, disclosure, additionality, verification, and the substantive quality of the financed activity.

Private and collaborative governance adds monitoring, information, reputation, and market-access mechanisms to this policy mix. Certification programs, zero-deforestation commitments, non-governmental monitoring, and supply-chain agreements can transmit environmental requirements beyond the direct reach of public enforcement. Evidence concerning their role in the decline in Amazon deforestation indicates that private coordination can complement public policy (NEPSTAD et al., 2014). Nevertheless, voluntary commitments tend to be selective in membership, commodity coverage, and territorial reach, while reputational sanctions depend on traceability and on the value that firms and consumers attach to compliance. Collaboration can improve information and collec-

tive action, but its effectiveness depends on the structure of relationships among actors and on whether those relationships match the ecological and institutional problem being addressed (BODIN, 2017). NGOs and private standards are therefore neither substitutes for the state nor merely symbolic actors; they are complementary governance mechanisms whose authority and capacity remain issue-specific.

Because these instruments address different margins, environmental governance is necessarily organized as a policy mix. Regulation can establish a binding floor, pricing can reveal heterogeneous abatement costs, payments can compensate conservation, finance can relax investment constraints, and private governance can improve traceability or market discipline. Their combination may be complementary, but it may also create overlap, contradiction, regulatory arbitrage, or unstable expectations. The effectiveness of a policy mix depends not only on the presence of individual instruments, but also on the consistency of its objectives, the coherence of its components, the credibility of implementation, and the sequencing of change (ROGGE; REICHARDT, 2016). Environmental policy under complex and uncertain conditions therefore requires portfolios of instruments and institutions rather than reliance on a universally dominant mechanism (STERNER et al., 2019). Institutional density should not be confused with institutional integration.

Four recurrent mismatches remain particularly relevant to the analytical problem developed in this article. The first concerns the *object* of governance. Existing instruments generally regulate an emission, a prohibited act, a contractual service, an eligible investment, or a disclosure obligation. Ecological functionality, however, is multidimensional and jointly affected by production, preservation, spatial interaction, and ecological feedback. Governing one observable component may improve that component without determining the condition of the wider ecological system.

The second mismatch is *spatial*. Property boundaries, municipal jurisdictions, national commitments, and spatially fungible markets do not necessarily coincide with watersheds, habitats, ecological corridors, climatic feedbacks, or other functional ecological units. Scale mismatches arise when responsibility is allocated at a level that does not correspond to the environmental process being governed, producing information loss, spillovers, and incomplete incentives (CUMMING et al., 2006). The literature on institutional fit similarly shows that sustainability depends on the correspondence among ecological processes, social organization, and governance arrangements rather than on decentralization or centralization in the abstract (EPSTEIN et al., 2015). Local governance can improve contextual information and connect benefits to affected agents, but cross-boundary effects still require coordination at broader levels.

The third mismatch is *temporal and distributive*. Returns from resource conver-

sion are often immediate and privately appropriable, whereas benefits from ecological maintenance may be delayed, uncertain, dispersed among beneficiaries, or reflected in avoided future losses. Legal obligations, payments, and finance can change parts of this comparison, but they do not automatically make the productive contribution of ecological functionality enter the agent's expected payoff. The fourth mismatch is *institutional*: formal objectives must pass through rules, budgets, information systems, administrative organizations, and enforcement practices before they affect behavior. Where translation, monitoring, or policy credibility is weak, the incentive perceived by the agent can differ substantially from the incentive intended by the rule.

Taken together, the evidence does not imply that existing environmental instruments are generally ineffective. Command-and-control regulation can deter illegal conversion; carbon pricing can reduce emissions; payments can alter land-use decisions; financial instruments can redirect capital; and collaborative governance can improve monitoring and coordination. Their performance, however, remains conditional on measurement, targeting, enforcement, institutional capacity, policy credibility, and fit with the relevant ecological process. More importantly for the present analysis, these mechanisms typically govern specific externalities, transactions, or legal obligations. Their coexistence does not by itself establish an integrated relationship in which ecological functionality jointly determines production, enters private payoffs, and is governed at the spatial scale of the ecological processes involved.

This residual disconnect defines the transition from context to theory. The analytical gap is not the absence of environmental concern from economic policy, nor does it justify rejecting regulation, markets, finance, or collective governance. It is the absence, for the purposes of this article, of a unified representation connecting the ecological state, productive outcomes, private incentives, and territorially aligned institutions. Section 3 therefore examines how the theories of externalities, property rights, common-pool resources, and natural capital illuminate separate components of this problem, before Section 4 reconstructs their connection through the proposed framework of economic commensalism.

3. Literature Review

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint

occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

Posicionar explicitamente em relação a Dasgupta (2021): O relatório The Economics of Biodiversity (Dasgupta Review) endogeneiza a natureza de forma sofisticada. Mostrar no que seu framework vai além desse trabalho já canônico.

4. Conceptual and Analytical Framework

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

; and (iv) to distinguish commensal resource use from predatory, extractive, and mutualistic configurations according to their joint effects on private viability and ecological functionality.

5. Theoretical Results

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

6. Discussion

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Sed do eiusmod tempor incididunt ut labore et dolore magna aliqua. Ut enim ad minim veniam, quis nostrud exercitation ullamco laboris nisi ut aliquip ex ea commodo consequat. Duis aute irure dolor in reprehenderit in voluptate velit esse cillum dolore eu fugiat nulla pariatur. Excepteur sint occaecat cupidatat non proident, sunt in culpa qui officia deserunt mollit anim id est laborum.

7. Final Remarks

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Proin vitae ligula tortor. Suspendisse erat orci, fermentum in consectetur id, hendrerit nec erat. Donec ac gravida urna. Fusce a est sed ligula congue posuere. Suspendisse dictum erat ut velit fringilla pellentesque. Pellentesque ante sem, accumsan tincidunt diam vitae, convallis aliquam dui. Nulla nec nulla nec dui ultrices tempor. Vivamus sit amet facilisis dolor, nec luctus tortor. Fusce tempor eget eros a bibendum. Donec sed dui vel dolor tempor iaculis. Duis laoreet risus nisl, eu maximus erat pellentesque sit amet. Morbi sodales metus sit amet lectus dictum mollis. Vestibulum nunc neque, tempus eu posuere eget, malesuada et massa. Nullam ullamcorper consequat diam quis cursus. Phasellus ultricies sollicitudin eros quis dictum. Duis molestie venenatis arcu eget fermentum.

Suspendisse condimentum sed nibh et bibendum. Etiam a placerat tellus. In id tellus purus. Pellentesque sagittis porttitor augue, nec commodo tellus facilisis eget. Phasellus mattis dictum pharetra. Nulla euismod, arcu ut pretium posuere, quam magna scelerisque ligula, ut blandit sem sem in eros. Sed rhoncus turpis vitae ex porttitor, a posuere velit aliquet. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Quisque at ligula molestie, ornare dui vitae, placerat lectus. Praesent porttitor neque quis finibus vulputate. Suspendisse potenti. Curabitur eu turpis ac elit dignissim venenatis. Nam porttitor ligula ac sollicitudin gravida. Donec leo magna, dapibus id finibus id, suscipit vitae lacus.

Suspendisse sit amet libero ac enim tempus cursus quis quis nisl. Nullam a risus vel augue eleifend iaculis. Proin vel posuere enim, accumsan congue massa. Nam venenatis, nibh non cursus auctor, quam ex congue tellus, in dictum arcu justo quis risus. Etiam sollicitudin, turpis ut convallis semper, leo dui elementum lectus, sit amet faucibus arcu ligula id quam. Cras nec justo sit amet risus mattis convallis. Duis cursus et urna quis malesuada. Maecenas facilisis porttitor felis, a ultrices dolor sodales vitae. Morbi eu consequat ante. Maecenas suscipit nec urna at finibus. Maecenas interdum rutrum mauris, sit amet viverra mauris finibus in. In mollis semper mauris, dictum tincidunt libero bibendum nec. Praesent sed felis non massa commodo molestie. Cras gravida odio et ante auctor auctor.

Morbi quis lacus fermentum, malesuada lectus at, bibendum nunc. Morbi mi odio, rutrum sed euismod vitae, tempus eget justo. Nullam faucibus lacus et sodales rutrum. Aliquam finibus augue id velit euismod, at maximus eros ullamcorper. In hac habitasse platea dictumst. Aenean vel maximus nunc. Nunc in ultricies ex. Mauris metus tortor, tempor nec tortor a, pretium hendrerit risus. Nulla sed porta risus, sed lobortis est. Nam lobortis velit porta feugiat ultrices. Phasellus sodales mauris et magna convallis sodales.

Proin tincidunt libero et lacus posuere porta. Vestibulum tellus ante, condimentum a diam vel, eleifend pellentesque lacus.

Sed scelerisque eros massa, id congue justo faucibus ac. Donec dolor nisi, faucibus euismod erat in, dictum iaculis mi. Duis nec venenatis mauris. Proin tincidunt nisl nec maximus egestas. Lorem ipsum dolor sit amet, consectetur adipiscing elit. Curabitur fringilla luctus nulla, nec faucibus sem pharetra sed. Ut venenatis odio eget fermentum aliquam. Sed ac ex velit. Vestibulum non euismod libero. Sed tincidunt turpis odio, id aliquet lorem pretium vel.

References

AZEVEDO, A. A. et al. Limits of brazil's forest code as a means to end illegal deforestation. **Proceedings of the National Academy of Sciences**, v. 114, n. 29, p. 7653–7658, 2017. Disponível em: <https://doi.org/10.1073/pnas.1604768114>. Acesso em: 25 aug. 2026.

BAYER, P.; AKLIN, M. The european union emissions trading system reduced CO2 emissions despite low prices. **Proceedings of the National Academy of Sciences**, v. 117, n. 16, p. 8804–8812, 2020. Disponível em: <https://doi.org/10.1073/pnas-.1918128117>. Acesso em: 25 aug. 2026.

BODIN, Ö. Collaborative environmental governance: Achieving collective action in social-ecological systems. **Science**, v. 357, n. 6352, p. eaan1114, 2017. Disponível em: <https://doi.org/10.1126/science.aan1114>. Acesso em: 25 aug. 2026.

BÖRNER, J. et al. The effectiveness of payments for environmental services. **World Development**, v. 96, p. 359–374, 2017. Disponível em: <https://doi.org/10.1016/j-.worlddev.2017.03.020>. Acesso em: 25 aug. 2026.

BÖRNER, J. et al. Mixing carrots and sticks to conserve forests in the brazilian amazon: A spatial probabilistic modeling approach. **PLOS ONE**, v. 10, n. 2, p. e0116846, 2015. Disponível em: <https://doi.org/10.1371/journal.pone.0116846>. Acesso em: 25 aug. 2026.

BRASIL. **Lei nº 12.651, de 25 de maio de 2012**. 2012. Diário Oficial da União: seção 1, Brasília, DF, ano CXLIX, n. 102, p. 1–8, 28 maio 2012. Dispõe sobre a proteção da vegetação nativa; altera diversas leis e revoga outras. Disponível em: <https://www.planalto.gov.br/ccivil_03/_ato2011-2014/2012/lei/l12651.htm>.

CAMPIGLIO, E. Beyond carbon pricing: The role of banking and monetary policy in financing the transition to a low-carbon economy. **Ecological Economics**, v. 121, p. 220–230, 2016. Disponível em: <https://doi.org/10.1016/j.ecolecon.2015.03.020>. Acesso em: 25 aug. 2026.

COASE, R. H. The problem of social cost. **Journal of Law and Economics**, v. 3, p. 1–44, 1960.

CUMMING, G. S. et al. Scale mismatches in social-ecological systems: Causes, consequences, and solutions. **Ecology and Society**, v. 11, n. 1, p. 14, 2006. Disponível em: <https://doi.org/10.5751/ES-01569-110114>. Acesso em: 25 aug. 2026.

DASGUPTA, P. **Human Well-Being and the Natural Environment**. 1st. ed. Oxford: Oxford University Press, 2001. ISBN 9780199247882.

DÖBBELING-HILDEBRANDT, N. et al. Systematic review and meta-analysis of ex-post evaluations on the effectiveness of carbon pricing. **Nature Communications**, v. 15, p. 4147, 2024. Disponível em: <https://doi.org/10.1038/s41467-024-48512-w>. Acesso em: 25 aug. 2026.

EPSTEIN, G. et al. Institutional fit and the sustainability of social-ecological systems. **Current Opinion in Environmental Sustainability**, v. 14, p. 34–40, 2015. Disponível em: <https://doi.org/10.1016/j.cosust.2015.03.005>. Acesso em: 25 aug. 2026.

FLAMMER, C. Corporate green bonds. **Journal of Financial Economics**, v. 142, n. 2, p. 499–516, 2021. Disponível em: <https://doi.org/10.1016/j.jfineco.2021.01.010>. Acesso em: 25 aug. 2026.

GOULDER, L. H.; PARRY, I. W. H. Instrument choice in environmental policy. **Review of Environmental Economics and Policy**, v. 2, n. 2, p. 152–174, 2008. Disponível em: <https://doi.org/10.1093/reep/ren005>. Acesso em: 25 aug. 2026.

HARDIN, G. The tragedy of the commons. **Science**, v. 162, n. 3859, p. 1243–1248, 1968.

INPE. **Como se monitora o desmatamento da Amazônia?** 2024. Disponível em: <https://www.gov.br/inpe/pt-br/acesso-a-informacao/perguntas-frequentes/principais-produtos-e-servicos-do-inpe/monitoramento-do-territorio-florestas/como-se-monitora-o-desmatamento>. Acesso em: 25 aug. 2026.

IPBES. **Summary for Policymakers of the Global Assessment Report on Biodiversity and Ecosystem Services of the Intergovernmental Science-Policy Platform on Biodiversity and Ecosystem Services**. Bonn, 2019. 56 p. Disponível em: <https://doi.org/10.5281/zenodo.3553458>. Acesso em: 25 aug. 2026.

IPCC. **Climate Change 2023: Synthesis Report. Contribution of Working Groups I, II and III to the Sixth Assessment Report of the Intergovernmental Panel on Climate Change**. Geneva, 2023. 184 p. Disponível em: <https://doi.org/10-.59327/IPCC/AR6-9789291691647>. Acesso em: 25 aug. 2026.

NEPSTAD, D. et al. Slowing amazon deforestation through public policy and interventions in beef and soy supply chains. **Science**, v. 344, n. 6188, p. 1118–1123, 2014. Disponível em: <https://doi.org/10.1126/science.1248525>. Acesso em: 25 aug. 2026.

OLIVEIRA, G. M. de; SCHNAIDER, P. S. B. Implementation of the brazilian forest code: A meso-institutional approach. **Journal of Institutional Economics**, v. 21, p. e26, 2025. Disponível em: <https://doi.org/10.1017/S1744137425100143>. Acesso em: 25 aug. 2026.

OSTROM, E. **Governing the Commons: The Evolution of Institutions for Collective Action**. Cambridge: Cambridge University Press, 1990.

PIGOU, A. C. The online library of liberty. **Online Library of Liberty**, 2010.

ROGGE, K. S.; REICHARDT, K. Policy mixes for sustainability transitions: An extended concept and framework for analysis. **Research Policy**, v. 45, n. 8, p. 1620–1635, 2016. Disponível em: <https://doi.org/10.1016/j.respol.2016.04.004>. Acesso em: 25 aug. 2026.

ROSENBLOOM, D. et al. Why carbon pricing is not sufficient to mitigate climate change—and how "sustainability transition policy" can help. **Proceedings of the National Academy of Sciences**, v. 117, n. 16, p. 8664–8668, 2020. Disponível em: <https://doi.org/10.1073/pnas.2004093117>. Acesso em: 25 aug. 2026.

SOARES-FILHO, B. et al. Cracking brazil's forest code. **Science**, v. 344, n. 6182, p. 363–364, 2014. Disponível em: <https://doi.org/10.1126/science.1246663>. Acesso em: 25 aug. 2026.

STERNER, T. et al. Policy design for the anthropocene. **Nature Sustainability**, v. 2, n. 1, p. 14–21, 2019. Disponível em: <https://doi.org/10.1038/s41893-018-0194-x>. Acesso em: 25 aug. 2026.

UNEP. **Emissions Gap Report 2025: Off Target**. Nairobi, 2025. Disponível em: <https://www.unep.org/resources/emissions-gap-report-2025>. Acesso em: 25 aug. 2026.

UNFCCC. **Kyoto Protocol to the United Nations Framework Convention on Climate Change**. 1997. Disponível em: <https://unfccc.int/process-and-meetings/the-kyoto-protocol>. Acesso em: 25 aug. 2026.

UNFCCC. **Paris Agreement**. 2015. Disponível em: <https://unfccc.int/process-and-meetings/the-paris-agreement>. Acesso em: 25 aug. 2026.

United Nations. **Convention on Biological Diversity**. 1992. Disponível em: <https://www.cbd.int/convention/text>. Acesso em: 25 aug. 2026.

WUNDER, S. et al. Payments for environmental services: Past performance and pending potentials. **Annual Review of Resource Economics**, v. 12, p. 209–234, 2020. Disponível em: <https://doi.org/10.1146/annurev-resource-100518-094206>. Acesso em: 25 aug. 2026.